

The basement membrane as a surface with a reservoir:

deformation, remeshing, tearing, and the Plexus2 container they live in

prototype/ecm/bm_ops.py · test_05_sheet.py · test_05b_plaque.py · test_05c_remesh.py · test_05e_conserve.py ·
log/okuda_ECM/05a-05c (05d-05f archived, see §4.2) · 10 August 2026

1. Mechanism

What the object is. A basement membrane is a sheet of crosslinked protein — collagen IV and laminin, with nidogen and perlecan — about 100 nm thick, that an epithelium secretes onto its own basal surface and then sits on. Every epithelium in the body has one. It is the boundary that makes a tissue a tissue: it separates epithelium from stroma, carries the mechanical load of the sheet above it, holds growth factors, passes solutes while stopping cells, and is the thing a tumour must breach to become invasive.

What it has to do, and why the two demands conflict. It must *hold together as a material*, which asks it to be stiff and continuous; and it must *keep covering a surface that grows*, which for the spheroid modelled here means an area rising by a factor of eleven. Those pull against each other, and biology resolves them with three mechanisms rather than with a compromise in stiffness:

- **it is replaced while it is stretched.** Collagen IV turns over on a half-life of 3–10 hours [10], comparable to the time in which the tissue grows, so the sheet enclosing a tripled radius is not the sheet that was there at the start.
- **it is attached, not merely adjacent.** Integrin clusters bind it to the cells beneath, transmitting traction both ways and letting the sheet slide as the tissue rearranges underneath.
- **it fails where it is not resupplied.** A hole is not a stress concentration on a fixed material; it is a place where removal beats deposition.

The equations those mechanisms are, before any discretisation. Each is a balance on a quantity carried per unit area of a deforming surface, and each has the same shape — a material derivative, a source, a sink, and a dilution term that appears because the surface itself is growing:

carried quantity	balance	what it decides
sheet material ρ	$D\rho/Dt = s - \rho/\tau_{\text{bm}} - \rho \dot{A}/A$	whether the sheet thins, holds, or tears
receptor n_f , bonds n_b	binding kinetics with a load-dependent off-rate	how hard it is held, and whether it slides
momentum	$\gamma \dot{\mathbf{x}} = \mathbf{f}$, massless and overdamped	where it sits and how fast it gets there

Why the mechanics is overdamped and not inertial. At the scale of a membrane in tissue fluid the Reynolds number is $\sim 10^{-10}$: inertia is not small, it is absent. The correct statement of motion is that velocity is proportional to force, and the consequence for everything downstream is that a *stiffness has no meaning on its own* — only the ratio k/γ , a relaxation rate, appears. The question a stiffness answers here is not “how strong” but “does the sheet relax faster than the tissue grows”.

What this becomes in Plexus2. A model is sets $\times$ fields $\times$ operators $\times$ schedule, and the biology above chooses all four. The sheet is a *surface*, so it is a node set (**bm**) with a hyperedge set of triangles over it (**bm_face**) that carries everything material; its thickness never becomes geometry, only a coefficient. Adhesion is a *relation between two entities*, so it is an edge set (**plaque**) whose single operator returns a force to both ends and cannot get them out of step. The receptor belongs to the *cell*, so it is a state on **cell**. And because the sheet gains and loses material, its *size is a state variable*: the sets are allocated with a reserve and carry the framework’s **occ** flag, exactly as MPM particles do for **cell_grow**.

One distinction is load-bearing throughout this note. **bm_refine** — changing the triangulation as the surface grows — is a *numerical* operation. It changes no physical quantity: not the surface, not the material state, not the mass. Secretion, turnover and proteolysis are *biological* mechanisms that change what the sheet is made of and how much there is. They share an implementation substrate, the **occ** reservoir, and nothing else. Conflating them is what lets a discretisation choice be read as a result — and the historical case is exactly that: the previous implementation of this sheet, as material points on a background grid, produced tearing that was set by the grid spacing rather than by any stress, on a sheet an eighth of a grid cell thick. §4 gives the gate that catches it and the measurement that passes it.

How the rest reads. §2 gives the equations from the coarsest statement down: the balance of forces on the surface, then the elasticity and the strain measure it needs, then the material budget, then the adhesion, then the

discretisation. §3 writes all of it as a Plexus2 container — sets, states, operators, schedule. §4 is the gate table: every claim in §2 with a threshold fixed before the run and the number that came back.

2. Equations and symbols

The whole model, in one line, before any of it is expanded. The sheet is a massless overdamped surface, so its equation of motion is a balance between the forces acting on it and the drag resisting them:

$$\gamma \dot{\mathbf{x}} = \underbrace{\mathbf{f}_{\text{el}}}_{\text{§2.1: it is elastic}} + \underbrace{\mathbf{f}_{\text{adh}}}_{\text{§2.3: it is attached}} + \underbrace{\mathbf{f}_{\text{con}}}_{\text{§2.3: it is outside the cells}}, \quad \mathbf{x} \leftarrow \mathbf{x} + \Delta t M \mathbf{f}, \quad M = 1/\gamma, \quad (1)$$

alongside two budgets on quantities the surface carries — its own material (§2.2) and the bonds holding it (§2.3) — and one statement about the mesh that carries none of it (§2.4).

The stability limit that follows, because it constrains every term above. With no mass there is no wave speed, so the limit is a *rate*:

$$\Delta t M \lambda_{\max}(\mathbf{H}) < 2, \quad (2)$$

$\mathbf{H}$ being the Hessian of the energy. This is about twenty times weaker than the CFL an inertial sheet would carry, which is why a membrane ten to a hundred times stiffer than the stroma does not set the timestep for the whole model. $\lambda_{\max}$ is *measured*, by power iteration on Hessian-vector products, and re-measured as the run proceeds: a St Venant–Kirchhoff membrane’s tangent stiffens as it stretches, and over 401 frames it grew $19.1 \times (4.56 \rightarrow 87.2)$, taking the substep count with it ($21 \rightarrow 194$). Fixing the substep count from the seeded value put the measured divergence threshold at 1.8 instead of 2, and the entire discrepancy was the Hessian having grown under the run’s own load.

2.1 The elastic surface, and the strain measure it needs

The first term of Eq. (1) is the sheet resisting being stretched. One constant-strain triangle per face, St Venant–Kirchhoff, thickness entering only as a multiplier:

$$W_f = A_f^0 \left[\mu \operatorname{tr}(\mathbf{E}_g^2) + \frac{\Lambda}{2} (\operatorname{tr} \mathbf{E}_g)^2 \right], \quad \mathbf{E}_g = \frac{1}{2} (\mathbf{C}_{\text{el}} - \mathbf{I}), \quad \mathbf{f} = -\frac{\partial}{\partial \mathbf{x}} \sum_f W_f, \quad (3)$$

$$Y_2 = E T \quad [\text{force per unit length}], \quad \mu = \frac{Y_2}{2(1 + \nu)}, \quad \Lambda = \frac{Y_2 \nu}{1 - \nu^2}. \quad (4)$$

The force is the exact gradient, taken by automatic differentiation, so the force and the energy cannot disagree — G4 checks it against a central finite difference and gets 5×10^{-10} relative.

Which needs a strain measure, and this is the one the whole note is written in. λ is the quantity every other statement here is written in, and it is a *per-triangle* number. For a face with nodes $\mathbf{x}_{0,1,2}$, take the two edge vectors from the first node now and in the reference configuration $\mathbf{X}$:

$$\mathbf{D}_s = [\mathbf{x}_1 - \mathbf{x}_0, \mathbf{x}_2 - \mathbf{x}_0] \in \mathbb{R}^{3 \times 2}, \quad \mathbf{D}_m = \begin{bmatrix} a & \mathbf{v} \cdot \hat{\mathbf{e}}_1 \\ 0 & \mathbf{v} \cdot \hat{\mathbf{e}}_2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad (5)$$

where $\mathbf{D}_m$ is the same pair written in the reference triangle’s *own* in-plane frame $(\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2)$, with $a = |\mathbf{X}_1 - \mathbf{X}_0|$ and $\mathbf{v} = \mathbf{X}_2 - \mathbf{X}_0$. Then

$$\mathbf{F} = \mathbf{D}_s \mathbf{D}_m^{-1} \in \mathbb{R}^{3 \times 2}, \quad \mathbf{C} = \mathbf{F}^\top \mathbf{F}, \quad \lambda_{1,2} = \sqrt{\operatorname{eig} \mathbf{C}}, \quad J = \lambda_1 \lambda_2 = \frac{\text{area now}}{\text{area then}}. \quad (6)$$

λ_1 is the larger principal stretch: 1 is unstretched. **This is what the movies colour by** — `lam` in `traj.npz`, `lam_geo` in `metrics.json`. Because $\mathbf{F}$ is read from where the nodes *are* against where they *were*, it is blind to what moved them: a node moved by a tether, a plaque, a grid or by hand is seen identically, and there is no path by which a displacement is applied and not counted. That is a property of Eq. (6) and not a result, which is why it can be certified in closed form (§4, G1–G4) before any physics runs.

There are two λ ’s and they must never be quoted as one. The reference metric is either frozen at seeding or allowed to creep (§4), giving

$$\lambda^{\text{geo}} = \sqrt{\operatorname{eig} \mathbf{C}} \quad (\text{against the SEEDED metric}), \quad \lambda^{\text{el}} = \sqrt{\operatorname{eig}(\mathbf{C}_0^{-1} \mathbf{C})} \quad (\text{against the REMEMBERED metric } \mathbf{C}_0). \quad (7)$$

λ^{geo} is what the tissue did to the sheet; λ^{el} is what the material feels. With no remodelling they coincide. With remodelling they do not, and the gap is the mechanism.

2.2 Memory, stiffening, supply

Remodelling is the reference metric creeping toward the current one, the 2D form of the archived per-edge $\dot{\ell}^0 = (L - \ell^0)/\tau_r$:

$$\mathbf{C}_0 \leftarrow \mathbf{C}_0 + \frac{\Delta t}{\tau_r}(\mathbf{C} - \mathbf{C}_0). \quad (8)$$

Collagen IV's half-life of 3–10 hours is 13–93 frames here, so this is resolved rather than instantaneous. A sheet enclosing a surface that triples in radius cannot do it elastically.

Strain stiffening, $E(\lambda) = E_0[1 + \beta(\lambda^{\text{el}} - 1)]$, applied explicitly from the previous step's stretch so that Eq. (3) remains the exact potential of the force. Candiello *et al.* measure $0.4 \rightarrow 3$ MPa on native basement membrane; **MPa is not a quantity this box has** — it is dimensionless and the stroma's modulus is the number 15 — so the falsifiable forms are the ratio $E(\lambda_{\text{max}})/E_0$ and the contrast $E_{\text{bm}}/E_{\text{ecm}} \in [10, 100]$.

Areal density, and which of the two is the state. The mass balance a growing sheet owes is usually written for ρ ,

$$\frac{D\rho}{Dt} = \underbrace{s(\mathbf{x}, t)}_{\text{secretion}} - \underbrace{\frac{\rho}{\tau_{\text{bm}}}}_{\text{turnover}} - \underbrace{\rho \frac{\dot{A}}{A}}_{\text{dilution}}, \quad s^* = \rho \left(\frac{1}{\tau_{\text{bm}}} + \frac{\dot{A}}{A} \right), \quad (9)$$

whose two removal terms are the same size here ($\sim 10^{-1} \text{ h}^{-1}$ each) — turnover is not a correction to growth, it is the same size as growth, which is why a sheet can accommodate growth at all.

But ρ is the wrong variable to store, and 05e is the run that changed it. Carry the *mass* m_f on each face and derive the density,

$$\rho_f = \frac{m_f}{A_f(\text{now})}, \quad \frac{dm_f}{dt} = s A_f - \frac{m_f}{\tau_{\text{bm}}}, \quad (10)$$

and three things follow that the ρ form does not give. The dilution term *disappears from the integration* — it is what dividing by a growing area does on its own, and Eq. (9) integrated alongside Eq. (10) would count it twice. $\sum_f m_f$ becomes a conserved quantity that only secretion and proteolysis may change, so a remesh can be *gated* on it (G18b, measured bit-identical). And run 128's failure mode becomes structurally impossible: mass is added to faces that already are the sheet, so there is no way to secrete material into the lumen.

With $s = 0$ the measured sheet thins to 1/11.1 of what it was seeded with, which is Eq. (10) with no source and an area $\times 11.1$. **That number is the baseline `bm_secrete` has to hold flat**, and it is the reason a fixed particle budget was never going to work.

2.3 Adhesion: a receptor on the cell, bonds on the edge, ligand on the sheet

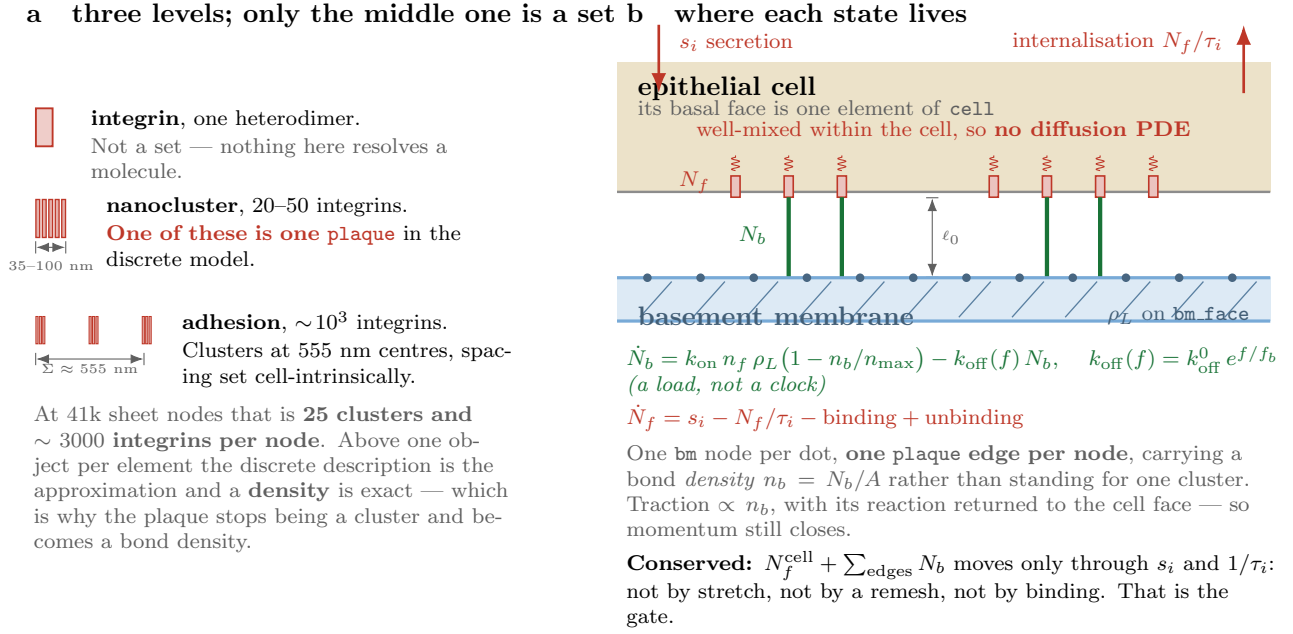


Figure 1: **(a)** An integrin is a heterodimer; 20–50 of them make a nanocluster 35–100 nm across [7]; clusters group into adhesions of $\sim 10^3$ integrins at a measured ~ 555 nm centre-to-centre spacing [8]. The discrete **plaque** of 05b was the MIDDLE level — one cluster — and at 41k sheet nodes there are 25 clusters and ~ 3000 integrins per node, so that description was resolving something the mesh cannot see. **(b)** The three states, each on the entity it belongs to. Free receptor is the CELL’s and is well mixed over its own basal face; bonds are a relation and live on the edge; ligand is sheet material and is already ρ .

A plaque is not an integrin; it is a cluster of them, and that hierarchy decides where every state goes. An integrin is a single transmembrane heterodimer. Integrins do not act alone — they assemble into *nanoclusters* of 20–50 molecules, 35–100 nm across, which is what lets an adhesion carry load no single bond could [7] — and those clusters group into an adhesion of order 10^3 integrins, spaced ~ 555 nm centre to centre by a cell-intrinsic mechanism rather than by the ligand [8].

The arithmetic then decides the representation, and it decides against discrete plaques. This sheet is 0.317 mm^2 at the last frame. At the measured spacing that is $\sim 1.03 \times 10^6$ clusters — against 40,962 nodes. **Twenty-five clusters and ~ 3000 integrins per node**, with a shot noise of 2%: above one object per element the discrete description is the approximation and the density is exact. 05b’s 1166 plaques were not a sparse sampling of adhesion, they were the wrong object by a factor of 883.

So the receptor becomes a field, and it belongs to the cell. An integrin is a protein in the *cell*’s basal plasma membrane; the sheet carries the *ligand*. Three states, each on the entity that owns it:

state	lives on	dynamics
N_f , free receptor	cell, its basal face	$\dot{N}_f = s_i - N_f/\tau_i - \text{binding} + \text{unbinding}$ — production and internalisation, no flux term (below)
N_b , bound bonds	edge plaque: cell→bm	Eq. (11): a bond has two ends, so it is a relation and not a state of either body
ρ_L , ligand	bm_face	laminin is sheet material; this is ρ , already there

And putting the receptor on the cell removes the PDE rather than adding one. The plasma membrane is continuous within a cell and interrupted at cell–cell junctions, so integrins do not diffuse between cells — lateral transport between them is exocytosis and endocytosis, a per-cell source and sink, not a flux. *Within* one cell, at $D \approx 0.1 \mu\text{m}^2/\text{s}$ over a 600 s frame, a receptor explores $\sqrt{4Dt} = 15.5 \mu\text{m}$ against a $\sim 10 \mu\text{m}$ cell: the basal face is **well mixed at our timestep**. So there is no Laplacian, no $\Delta t D/h^2 < \frac{1}{2}$ constraint — which would have cost ~ 14 substeps and, unlike the sheet’s rate bound, would have *tightened* at every refinement — and the receptor obeys a per-cell ODE. The assumption is stated rather than silent: if a cell is ever resolved by more than one face, the diffusion term returns.

The bonds themselves are the classical clutch, with Bell’s force-dependent unbinding:

$$\dot{N}_b = k_{\text{on}} n_f \rho_L \left(1 - \frac{n_b}{n_{\text{max}}}\right) A - k_{\text{off}}(f) N_b, \quad k_{\text{off}}(f) = k_{\text{off}}^0 e^{f/f_b}, \quad n_b = \frac{N_b}{A}, \quad (11)$$

and the traction on the sheet is $\propto n_b$, with the reaction returned barycentrically to the cell’s face exactly as `plaque_pull` does now — so G9 and G10 keep passing unchanged. **The plaque edge set survives and is reinterpreted:** one edge per sheet node, taken from the `bm_contact` map, carrying a bond *density* for its patch rather than standing for one cluster.

The force those bonds exert, and one sign that cost a run. The normal law is a spring at the linkage’s own rest length, scaled by how many bonds the patch is carrying, and the tangential law is friction against the relative velocity of sheet and cell:

$$\mathbf{f}_n = -n_b \kappa_n (d - \ell_0) \hat{\mathbf{n}}, \quad d = (\mathbf{x}^{\text{bm}} - \mathbf{p}) \cdot \hat{\mathbf{n}}, \quad \mathbf{f}_k^{\text{epi}} = -w_k (\mathbf{f}_n + \mathbf{f}_t), \quad (12)$$

$$\dot{\mathbf{x}}_{\parallel} = \frac{\dot{\mathbf{x}}_{\parallel}^{\text{prov}} + M \xi \mathbf{v}_{\parallel}^{\text{epi}}}{1 + M \xi}, \quad \xi = n_b \kappa_b / k_{\text{off}}, \quad \mathbf{f}_t = \xi (\mathbf{v}_{\parallel}^{\text{epi}} - \dot{\mathbf{x}}_{\parallel}). \quad (13)$$

$\hat{\mathbf{n}}$ is the epithelial face’s *outward* normal, not the line joining the two points, and that is a correctness fix rather than a refinement: along the line, the equilibrium $|\mathbf{x}^{\text{bm}} - \mathbf{p}| = \ell_0$ is satisfied on *either* side, so a sheet that has sunk through the epithelium sits at the same energy as one resting on it. 05e measured the consequence — a sheet that started 5.96×10^{-4} outside (which is ℓ_0 , correct) and ended 7.7×10^{-3} *inside*, thirteen ℓ_0 through the cell layer — while the standoff metric, being an unsigned distance, reported $+4.0 \times 10^{-3}$ throughout. **An unsigned quantity cannot answer a question about a side.**

Eq. (13) is *solved*, not *evaluated*: an explicit $-\xi \mathbf{v}$ is a stiffness of $\xi/\Delta t$, which *grows* as the substep shrinks, so refining the step to stabilise it makes it worse. And a signed adhesion is still not sufficient — with the sign fixed the sheet ended 89.6% inside, because the missing statement is one no adhesion model contains:

$$\mathbf{f}_i^{\text{contact}} = -k_c \min(d_i - \ell_0, 0) \hat{\mathbf{n}}_i, \quad \text{one-sided, at every node.} \quad (14)$$

`bm_contact` says that a basement membrane cannot be *inside* the cells. It holds at every node and not only where adhesion happens to be, it pushes out and never pulls in, and its reaction goes to the three vertices of the epithelial face so the momentum gate still covers it. It moved the signed standoff from -9.87×10^{-4} to $+1.97 \times 10^{-6}$ and the fraction of nodes on the wrong side from 89.6% to **0.0%**. This is what the archived `cell_exclude_3d` did and got wrong: that operator *projected* particles out, repositioning them without touching $\mathbf{F}$, which launders deformation — run 88’s sheet was carried outward as a decal at a strain of 7×10^{-4} against a true stretch of 3.4. A force cannot do that.

Three defects close at once, which is why this is a change of scheme and not an elaboration. The standoff (G12) was $3.43 \ell_0$ because 96% of nodes were unanchored and the sheet sagged between plaques; a distributed traction has nothing to sag between. Slip (G13) was unmeasurable because a fixed barycentric anchor is a tangential pin — slip *is* bond turnover, the sheet advancing as k_{off} releases bonds that rebind ahead, and ξ stops being a coefficient and becomes $n_b \kappa_b / k_{\text{off}}$. And rupture was a clock because a fixed threshold on a monotonically rising load is reached by everyone eventually; $k_{\text{off}}(f)$ gives a *steady-state* bound fraction that depends on load, so “fraction bound versus stress” becomes a curve.

What is conserved, and therefore what is gated: $N_f^{\text{cell}} + \sum_{\text{edges}} N_b$ changes only through s_i and $1/\tau_i$ — not by stretch, not by a remesh, not by binding, which merely moves receptor from one column to the other. That is the G18b treatment applied to receptor instead of mass.

The discrete plaque is not deleted; it is demoted to a refined patch. Where individual rupture and turnover matter — a crack tip, an invadopodium — a locally refined mesh can resolve Σ and carry discrete clusters again. The gate on that is a convergence test: the discrete model at high density must reproduce the density model.

2.4 The reservoir: a sheet whose size is a state variable

This is the part the archived line had no equivalent of, and it is what makes the rest possible.

What a remesh is, and what it must never be. `bm_refine` is a numerical topology update. The physical surface is unchanged — the same points of space, the same enclosed volume, the same material — and so is everything the material knows about itself. It exists because a fixed triangulation stops resolving a surface whose area grows by an order of magnitude, and for no other reason. **Nothing biological may be inferred from a refinement, and no biological operator may be implemented as one.** A tear that appears because

the mesh was coarse, a thinning that appears because triangles grew, a stiffening that appears because elements changed shape — each of those is a discretisation artefact wearing a mechanism’s name, and the MPM sheet whose separation was set by Δx is the cautionary case.

What must be carried across a remesh, unchanged or conservatively. Three things, and each is gated:

carried	how	why it is load-bearing
$\mathbf{C}_0$, the remodelling state	inherited identically ($\mathbf{C}_0$ is intensive and lives in material coordinates)	it is the sheet’s memory of what it has forgotten; resetting it makes a refined sheet elastically virgin
m_f , the mass	extensive: each child takes a quarter, so $\sum_f m_f$ is invariant (measured bit-identical)	$\rho = m_f/A_f$ is then derived, so a split that does not conserve mass would be a secretion nobody asked for
plaque attachments	node indices survive (nodes are appended, never removed); the plaques are RE-SEEDED to hold Σ^{-2}	measured : held at 0.996 of Σ^{-2} , against 0.145 for the fixed-fraction rule 05b used

Both of the last two rows were defects when this section was first written, and 05e is the run that fixed them. The plaque one is worth a number: 05b’s rule — anchor a fixed fraction of the nodes — did not lose a factor of 4 per refinement as predicted, it lost **a factor of 6.9 over the run**, because the fraction is wrong twice over: once at each split, and continuously as the area grows between them. `plaque_seed` now takes Σ in units of T and tops the count up to $A(t)/\Sigma^2$ every frame, which is also the biology — a growing epithelium makes more hemidesmosomes, it does not stretch the ones it has. Every plaque number in `05b_plaque` is therefore a number at the wrong density, and its density sweep reads as three arbitrary densities rather than as 100% / 20% / the measured one.

The problem, as a number. With a fixed mesh the sheet’s area goes $\times 11.1$ over 401 frames on the same 5120 triangles: the mean edge length goes $0.00305 \rightarrow 0.0102$ box units, $3.4\times$ its seeded value, and each triangle covers eleven times the tissue it started on. A mass balance cannot add material to a set whose size is fixed, and a proteolytic operator cannot remove it.

The pattern already exists in the framework. `engine.py:453` allocates `grow_reserve` dormant MPM particles at `occ = 0` for `cell_grow` to wake, and `engine.py:134` counts a set’s live members as (`occ > 0`). The sheet now carries the same flag on *both* of its sets. Nodes and faces are allocated to their maximum at seeding — 40,962 node slots and 81,920 face slots for a sheet that starts at 2562 and 5120 — and every operator that changes the sheet’s size flips `occ` rather than reallocating. Three consequences, and the third is the reason to do it this way rather than by growing arrays:

1. `bm_refine` wakes three faces per parent and one node per edge;
2. `bm_degrade` puts faces to sleep, and a hole is then simply a region with no live faces;
3. **an index survives all of it.** A plaque holds a *node* index, and nodes are only ever appended and never removed, so no adhesion is invalidated by a refinement or by a tear. Had the plaque bound to a face, every one of them would need reconciling after every split.

Refinement is a global $1 \rightarrow 4$ midpoint split of every live face, triggered when the mean live edge length passes a multiple of its seeded value. It is conforming by construction — every edge is split, so no face is left with a hanging node on one of its sides — and needs no red–green closure. It is a *material* trigger, not a schedule: a sheet that is not being stretched never refines.

The trap, and the identity that avoids it. A child triangle whose reference frame is rebuilt from its *current* shape reports $\lambda = 1$ the moment it is born, so a sheet that refines as it grows would silently forget everything it had been stretched by — the mesh version of the laundering that made run 130 report 13% of its true stretch. The four children of a midpoint split are related to their parent by a *constant* map $\mathbf{S}_k$ in material coordinates, so

$$\mathbf{D}_m^{\text{child}} = \mathbf{D}_m^{\text{parent}} \mathbf{S}_k \implies \mathbf{D}_m^{-1, \text{child}} = \mathbf{S}_k^{-1} \mathbf{D}_m^{-1, \text{parent}}, \quad A^0 \rightarrow \frac{A^0}{4}, \quad \mathbf{C}_0, \rho, Y_2 \text{ inherited}, \quad (15)$$

with, for children $(\mathbf{v}_0, \mathbf{m}_{01}, \mathbf{m}_{20})$, $(\mathbf{v}_1, \mathbf{m}_{12}, \mathbf{m}_{01})$, $(\mathbf{v}_2, \mathbf{m}_{20}, \mathbf{m}_{12})$, $(\mathbf{m}_{01}, \mathbf{m}_{12}, \mathbf{m}_{20})$,

$$\mathbf{S}_{1..4} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \frac{1}{2} \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}, \quad \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, \quad \frac{1}{2} \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}, \quad \det \mathbf{S}_k = \frac{1}{4} > 0 \quad \forall k. \quad (16)$$

Every $\det \mathbf{S}_k$ is positive, so orientation is inherited too. Substituting Eq. (15) into Eq. (6) gives $\mathbf{F}^{\text{child}} = \mathbf{F}^{\text{parent}}$ *identically*, and that identity is gate G15: measured at 2.3×10^{-14} on a loaded sheet, with the total area and the total elastic energy **bit-identical** across the split. Refinement is not approximately invisible to the mechanics; it is exactly invisible.

One deliberate omission. The midpoints are placed on the *chord* and are not projected onto the surface. Projecting would smooth the sheet and change λ , breaking Eq. (15) by construction: the smoothing has to come from the dynamics, not from the remesher.

Tearing is the same mechanism with the flag going the other way. A face dies when a criterion is met — λ^{el} past a break strain, the second Piola stress past a yield, or (the version the biology prefers) ρ below a critical areal density, which couples the tear to Eq. (9): a basement membrane fails where it is not resupplied. The rim of the hole is free, no node is removed, and the mesh stays conforming: opening a polar cap of 106 faces produced 34 rim edges and zero edges carrying three faces.

Why this needs a gate that MPM’s tearing never had. LADDER.md:24 flagged it at the outset: “MPM separates material that thins past the grid’s support, so tearing may be set by dx rather than by stress. A tear that moves when the grid is refined is numerical.” The refinement cross-check (run 90, $n_{\text{grid}} 64 \rightarrow 128$) was planned and never ran, on a sheet 1/8 of a grid cell thick that was carrying the stroma’s **youngs** 15 rather than the 400 its spec declared. So the visually convincing MPM tearing is, as of today, untested; G19 in §8 is that test.

2.5 Symbols

Geometry and strain — Eqs. (5)–(7)		
$\mathbf{x}_i$	node position	one corner of a triangle; the state of set bm
$\mathbf{D}_s, \mathbf{D}_m$	current, reference edge matrices	of one face; $\mathbf{D}_m$ is in that face’s own 2D frame
$\mathbf{F}$	deformation gradient	3×2 : maps the reference triangle to the current one
$\mathbf{C}, \mathbf{C}_0$	current, remembered metric	$\mathbf{C} = \mathbf{F}^\top \mathbf{F}$; $\mathbf{C}_0$ starts at $\mathbf{I}$ and creeps
λ_1, λ_2	principal stretches	1 = unstretched. <i>Per face</i> , never per node
$\lambda^{\text{geo}}, \lambda^{\text{el}}$	geometric, elastic stretch	what the tissue did / what the material feels
J	areal ratio	$\lambda_1 \lambda_2$
A_f^0	reference area	of face f
Material — Eqs. (3)–(4), (8)–(9)		
E	Young’s modulus	force per unit <i>area</i> ; meaningless here without T
T	thickness	enters ONLY as a multiplier; never resolved as geometry
$Y_2 = ET$	2D stretch modulus	force per unit <i>length</i> — the sheet’s actual stiffness
$\nu; \mu, \Lambda$	Poisson ratio; 2D Lamé	in the plane of the sheet
β	stiffening coefficient	$E(\lambda) = E_0[1 + \beta(\lambda - 1)]$
τ_r	remodelling time	how fast the sheet forgets being stretched, IN FRAMES
$\rho, \tau_{\text{bm}}, s$	areal density, turnover time, secretion rate	of collagen IV, Eq. (9)
Integration — Eq. (2)		
$M = 1/\gamma$	mobility	force to velocity; the sheet has no mass, so only M appears
$\lambda_{\text{max}}(\mathbf{H})$	largest Hessian eigenvalue	MEASURED, every ten frames; it grew 19.1× over a run
$s = \Delta t M \lambda_{\text{max}}$	stability group	the bound is $s < 2$; the substep count is set to hold it
ζ	relaxation e-folds per frame	does the sheet relax faster than the tissue grows
Reservoir — Eqs. (15)–(16)		
occ	occupancy flag	1 = live, 0 = dormant. On bm AND on bm_face
S_k	child map	constant 2×2 in material coordinates; $\det = \frac{1}{4}$
m_{ij}	midpoint node	of the edge between nodes i and j , placed on the CHORD
Plaque — Eqs. (12)–(13)		
ℓ, ℓ_0	current, rest length	of one plaque; $\ell_0 = 0.3 T$
κ_n	normal stiffness	of the plaque (not the sheet’s Y_2 , not the tether’s κ)
ξ	tangential friction	against RELATIVE velocity of sheet and epithelium
w_k	barycentric weights	of the attachment point in its epithelial face; also the reaction’s weights
N_f	free receptor	integrins on ONE CELL ’s basal face; a number, not a density
N_b, n_b	bound bonds, bond density	on one plaque edge; $n_b = N_b/A$. The traction is $\propto n_b$
ρ_L	ligand density	laminin on the sheet; this is ρ of §2.2 under its binding name
$k_{\text{on}}, k_{\text{off}}^0, f_b$	binding rate, unloaded off-rate, Bell force	$k_{\text{off}}^0 = k_{\text{off}}^0 e^{f/f_b}$: load shortens a bond’s life
n_{max}	saturation	the most bonds a patch can carry
s_i, τ_i	receptor production, internalisation time	of the CELL , not of the sheet
Σ	cluster spacing	centre-to-centre between integrin CLUSTERS , not between integrins; ~ 555 nm measured. Now sets n_{max} , no longer a plaque COUNT
κ	tether stiffness	05a only: the one-sided spring the plaque replaces

3. The Plexus2 container

Two questions decide whether any of the above can be written down: what the *sets* are, and what kind of operator each mechanism is.

Sets, and one structural claim. A triangle is not a body: it is a *relation among three nodes*, exactly as a

junction is a relation between two vertices in the epithelium. So the sheet is two sets, not one — **bm** carries position and **bm_face** carries everything material — and **bm_face** is a hyperedge set of arity 3 with **.a**, **.b**, **.c** index buffers, the same shape as **junction** $\subseteq \text{vertex}^2$ and **plaque** $\subseteq \text{cell} \times \text{bm}$. Putting the material state on the faces rather than on the nodes is what makes Eq. (15) a local inheritance rule instead of an interpolation.

set	kind	one element is	state it carries
sheet	node, depth 0	the membrane as an organ	total area, $\langle \rho \rangle$, live counts
bm	node, depth 1	one node of the membrane	pos , vel , occ
bm_face	hyperedge $\subseteq \text{bm}^3$	one patch of membrane	$\mathbf{D}_m^{-1}$, $\mathbf{C}_0$, A^0 , ρ , Y_2 , age, occ
cell	node (the epithelium)	one epithelial cell	(<i>its own state</i>) plus N_f , free integrin on its basal face
plaque	edge cell $\rightarrow$ bm	one patch of adhesion, ONE PER SHEET NODE — not one cluster	N_b bound bonds, ℓ_0 , κ_n , barycentric w_k , load
fibril	edge bm $\rightarrow$ ecm	one anchoring fibril	rest length, load
protease	field	MMP concentration	scalar, diffusing (drives bm_degrade)

Operators, in Plexus2’s taxonomy: *Seed* creates a set, *Lateral* moves state along an edge set, *Exchange* scatters to or gathers from a field, *Rewire* changes which elements are connected, and *Divide/Die* change how many exist. The reservoir is what lets the last two be $O(1)$ flips.

operator	kind	acts on	mechanism	its gate
bm_seed	Seed	bm, bm_face	lay the sheet on the recorded surface; allocate the reservoir	frame 0 reports $\lambda = 1$
bm_elastic	Lateral	bm_face	Eqs. (3)–(4), force as the exact gradient	G1–G5
bm_stiffness	Lateral	bm_face	$E(\lambda) = E_0[1 + \beta(\lambda - 1)]$	E/E_0 ratio, not MPa
bm_remodel	Lateral	bm_face	Eq. (8)	$\lambda^{\text{el}} \rightarrow 1$ while λ^{geo} holds
bm_refine	Divide	bm_face	<i>numerical maintenance only</i> : Eqs. (15)–(16), wake 3 faces + 1 node per edge	G14–G18c
bm_secrete	Lateral	bm_face	Eq. (10): add MASS to existing faces [10]	G24–G26
bm_degrade	Die	bm_face	proteolysis, or ρ below critical [11]	G19–G23
integrin_turnover	Lateral	cell	$\dot{N}_f = s_i - N_f/\tau_i$: the cell makes and recycles its own receptor. No flux term — the basal face is well mixed	receptor conserved up to s_i , $1/\tau_i$
plaque_bind	Lateral	plaque	Eq. (11), with Bell’s $k_{\text{off}}(f)$ [6]	bound fraction vs LOAD, not vs frame
plaque_seed	Rewire	plaque	one edge per sheet node, from the bm_contact map; rebuilt on refinement	one edge per live node, always
plaque_pull	Lateral	plaque	Eqs. (12)–(13), <i>with its reaction deleted</i> : rupture is now $k_{\text{off}}(f)$ in Eq. (11), a rate, not a threshold	G9–G13
plaque_rupture	—	—	Eq. (14): the cells exclude the sheet, one-sided, at every node	—
bm_contact	Lateral	bm, cell	the stroma is loaded <i>through</i> the sheet [13]	G28: the sheet stays outside
fibril_pull	Lateral	fibril	pores pass solutes, stop cells [12]	stroma displacement, sheet present vs. absent
bm_barrier	Exchange	bm_face, fields		flux across an intact vs. breached sheet

Schedule, and the division is not cosmetic. Anything that carries a force runs *inside* the substep block, at the sheet’s own rate; anything that changes the sheet’s topology or its mass runs once per frame, because a set whose membership changed mid-substep would be integrated against two different index sets in one step:

```

schedule:
- bm_refine          # frame level: topology. Trigger = mean live edge > 1.45 x seeded
- bm_secrete         # frame level: mass.      Eq. (10), wakes faces
- bm_degrade         # frame level: mass.      Eq. (10) / stress, sleeps faces
- plaque_seed        # frame level: rewire.    turnover, which is what permits slip
- {substep_dt: 4.0e-4, steps: [bm_stiffness, bm_elastic, plaque_pull, fibril_pull,
                                mpm_strain, mpm_scatter, mpm_grid_update, mpm_gather]}
- bm_remodel         # frame level: Eq. (9), tau_r in frames
- plaque_rupture     # frame level: rewire on the load the substeps just produced

```


The substep count is not a constant: it is set from the measured $\lambda_{\max}$ to hold $\Delta t M \lambda_{\max}$ at a chosen group below 2 (§3), which over a run means $21 \rightarrow 194$.

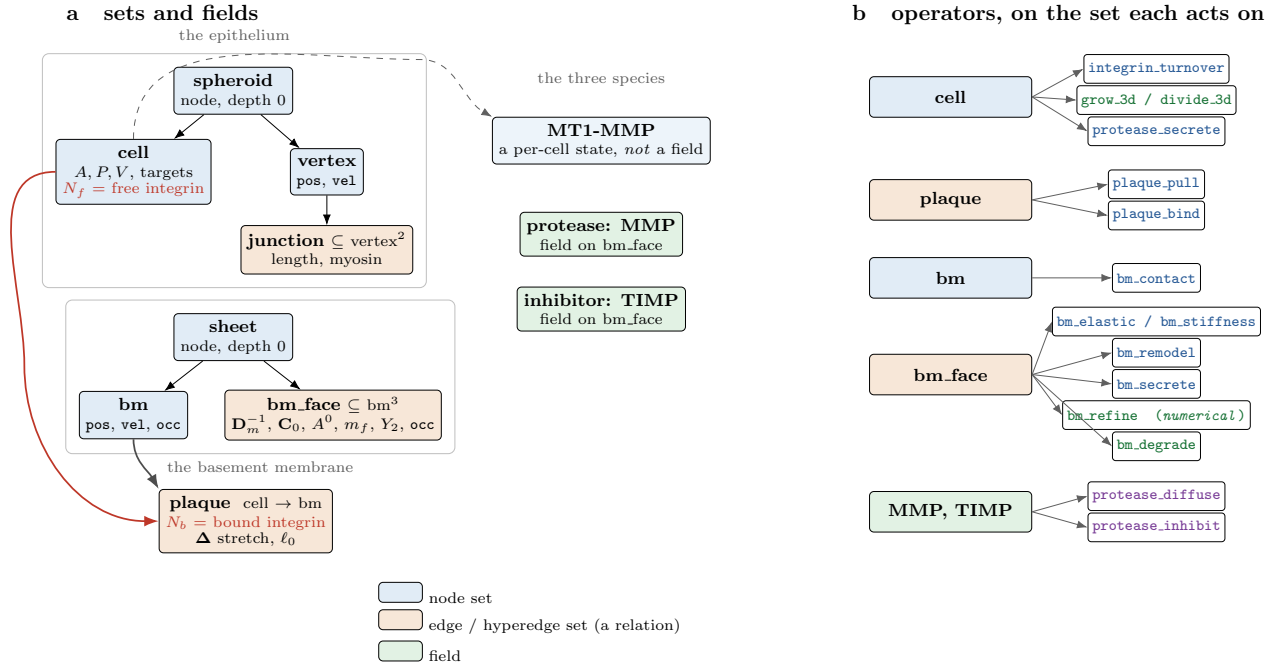


Figure 2: **(a)** Every set and field, and which kind each is. A node set carries positions; an edge or hyperedge set is a *relation* and carries state belonging to neither endpoint — a junction to neither cell, a plaque to neither the cell nor the sheet, a triangle to none of its three nodes. **(b)** Every operator, on the set it acts on, coloured by kind. Anything carrying a *force* sits on a set whose elements have positions; anything carrying *mass* or *number* sits on the set that owns the quantity. **bm_refine** is the only operator here that is numerical rather than biological.

3.1 The two diffusible species, and the one that is not

Proteolysis introduces the first *fields* in this model, and deciding which of the three species deserves to be one is not a modelling preference — it is arithmetic.

species	what it is	where it lives	why
MMP	soluble MMP-2/9, ~72 kDa, cuts collagen IV	field on bm_face	it diffuses freely, $D \sim 10\text{--}100 \mu\text{m}^2/\text{s}$
TIMP	soluble inhibitor, binds MMP 1:1	field on bm_face	likewise — and it is the <i>sink</i>
MT1-MMP	membrane-type MMP14, transmembrane	a per-cell <i>state</i>	it never enters the extracellular space

A soluble protease cannot open a hole, and the arithmetic says so before any run. In one 600 s frame a soluble MMP travels $\sqrt{4Dt} = 155\text{--}490 \mu\text{m}$, against a spheroid $318 \mu\text{m}$ across and a cell of $10 \mu\text{m}$. It is therefore *uniform over the entire sheet within a single frame*, and a uniform protease dissolves rather than perforates — the same failure uniform starvation produced, for the same reason. Localising to one cell would need $D < 0.042 \mu\text{m}^2/\text{s}$, $240\text{--}2400\times$ slower than any protein of that size. That is not a parameter range, it is a different molecule, and the different molecule is MT1-MMP: *tethered*, acting where its cell put it. This is why focal basement-membrane breach is done by membrane-type MMPs and invadopodia rather than by soluble enzyme.

What makes the soluble field worth solving at all is the sink. With $Dc/Dt = D\nabla_s^2 c + s - kc$ the source decays over $\sqrt{D/k}$, so the field acquires a *length* instead of filling the sphere — and k here is not a fitted decay but the reaction with TIMP. The breach diameter becomes a *prediction* of the MMP/TIMP balance: a $10 \mu\text{m}$ breach needs an effective k of $0.1\text{--}0.5 \text{s}^{-1}$ (half-life 1.4–6.9 s), a $50 \mu\text{m}$ breach $0.004\text{--}0.02 \text{s}^{-1}$ (35–170 s). A number to check, not to choose.

Three new operators, and one costs what the others do not. **protease_secrete** (Lateral, on **cell**) is the source; **protease_inhibit** (Exchange) is the 1:1 reaction; **protease_diffuse** (Exchange) is the only operator in this model carrying a *spatial* coupling, and so the only one whose stability limit involves the mesh. Explicit diffusion needs $\Delta t D/h^2 < \frac{1}{2}$ — at these D , 1,300–13,400 substeps per frame, *quadrupling at every 1→4 refinement*, unlike the sheet’s own bound which is a rate and mesh-independent. It is therefore solved semi-implicitly, $(\mathbf{I} + \Delta t D \mathbf{L})c = c_{\text{old}}$ by conjugate gradient on a finite-volume face Laplacian: unconditionally stable, one solve per frame, indifferent to h .

The same argument applied to the integrin gives the opposite answer for the opposite reason. A receptor is well mixed because it is *confined* to one cell's basal face ($\sqrt{4Dt} = 15.5 \mu\text{m}$ over a $10 \mu\text{m}$ cell) and needs no field; a soluble protease would be well mixed because it *travels far*. **Confinement, not diffusivity, decides whether a species is worth resolving as a field.**

Proteolysis, and the mechanism the literature forced

What the proteases are for, before any of their own chemistry. Everything up to here builds a barrier: a sheet that carries load, holds its density, and stays attached. The proteases are the only operators in the model that *remove* it, and they do so *locally* — so they are the mechanism of **barrier breach**, the step by which an epithelium stops being enclosed and a cell can cross into the stroma. Their target is the sheet's material (**bm_degrade** acts on **bm_face** mass); adhesion is affected only downstream, because a bond needs ligand, so ρ falling takes n_b with it.

05g_degrade put the three species on the entities the confinement argument assigns them and measured which one can make a hole. The answer was unambiguous and one-sided:

arm	faces dead	under MT1	elsewhere	sheet breached
both arms	317	317	0	6.2%
soluble only	0	0	0	0%
tethered only	317	317	0	6.2%
no protease	0	0	0	0%
no inhibitor	5120	317	4803	100%

The last row is the argument of §3.1 as an experiment: **remove TIMP and the sheet dissolves entirely**, with 4803 of the deaths nowhere near a cell that expresses MT1-MMP. With the inhibitor present the diffusible arm kills nothing at all and the tethered arm kills exactly its own footprint. A soluble protease has no length scale unless something quenches it.

Two defects that the runs found rather than the design. First, neither field had a *clearance* term, so TIMP accumulated without bound (0.0318 at frame 20, 0.0438 at frame 25) — its only sink was a 1:1 reaction with an MMP twenty times scarcer, so there was no steady state for the pair to reach. A first-order clearance is not a refinement: the $\sqrt{D/k}$ that justifies solving the field *is* that k . With it, both fields plateau and drift by $< 10^{-4}$ over twenty frames. Second, a five-point sweep of the MMP:TIMP source ratio over a hundredfold range returned *identically* 317 dead at every point, because the direct tethered cut had been set $350\times$ larger than anything the soluble arm could contribute — the sweep axis had no effect on the answer, and the holes were the source stencil in every case.

TIMP-2 is the adaptor as well as the inhibitor, and that predicts a bell

The pure-inhibitor law of **05g** is half of the mechanism. In the canonical model [1] TIMP-2 is the *bridging adaptor* that makes activation possible: its C-terminus tethers proMMP-2 and its N-terminus binds MT1-MMP, and the prodomain is then cleaved by a *second*, *TIMP-2-free* MT1-MMP [2]. TIMP-2 therefore both presents the zymogen and poisons the enzyme that must cut it, and the dose response is biphasic. The dissection is clean: an artificial receptor that tethers proMMP-2 *without* inhibiting MT1-MMP has no descending limb [3], so the fall-off is inhibition and not saturation.

With T the tethered enzyme and K the MT1-TIMP dissociation constant, occupancy is Langmuir and the rate needs one occupied site and one free one:

$$[\text{MT1}\cdot\text{TIMP}] = T \frac{c_T/K}{1 + c_T/K}, \quad [\text{MT1}_{\text{free}}] = \frac{T}{1 + c_T/K}, \quad \text{act} \propto T^2 c_{\text{pro}} \frac{c_T/K}{(1 + c_T/K)^2}, \quad (17)$$

which is bell-shaped and **peaks at** $c_T = K$. Both the shape and the location are therefore predictions, and **05g's** pure-inhibitor law cannot produce a peak at all — it can only decay — so one sweep distinguishes them and **05g** is the control.

Measured (**05h_ternary**, eight TIMP levels, four species): the activation rate rises and falls, $4.5 \times 10^{-2} \rightarrow 6.2 \times 10^{-2} \rightarrow 2.4 \times 10^{-2} \rightarrow 7.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-3}$, and the pure-inhibitor control over the same sweep is monotone decreasing, as Eq. (17) requires of it. Two arrangements had to be corrected before the curve could be read at all, and both are worth recording because they are measurement errors rather than modelling ones: the activation had to be made *rate-limiting* (at a large k_{act} the whole zymogen pool converts every frame whatever TIMP does, and saturation erases the dose response); and the rate has to be read *while the substrate still exists*, since at the TIMP levels where the mechanism works best it destroys every MT1-bearing face and a late-window average then reports exactly zero at the peak.

4. Gates

A gate is a number with a threshold decided *before* the run and a stated consequence if it fails. G1–G4 and G14–G16 run before any physics, as `bm_ops.selftest()`; a rig refuses to start if they fail.

gate		threshold
<i>the strain measure (§2)</i>		
G1	rigid motion leaves $\lambda = 1$	$< 7.5 \times 10^{-9}$ 10^{-6}
G2	dilation by s : $\lambda_1 = \lambda_2 = s$, every face	$< 2.5 \times 10^{-8}$ (J : 1.1×10^{-13}) 10^{-6}
G3	affine map gives the singular values of $\mathbf{A}$	$< 8.9 \times 10^{-15}$ 10^{-10}
G4	force $= -\partial W / \partial \mathbf{x}$, vs finite difference	$< 5 \times 10^{-10}$ 10^{-6} rel.
<i>the sheet under load (05a_sheet)</i>		
G5	stretch fidelity vs the applied $R(t)/R(0)$	> 0.9935 (MPM: 0.13, 0.98) 0.95
G6	control $\kappa = 0$: nothing pulls, nothing stretches	$= 1.00000$ 1.00000
G7	control, rigid drive: motion is not deformation	$= 1.00000$ 1.00000
G8	divergence at $\Delta t M \lambda_{\max} = 2$	stable ran at 1.9, NaN at 2.1 < 2 , NaN > 2
<i>the plaque (05b_plaque)</i>		
G9	momentum, bookkeeping: $ \sum \mathbf{f}_{\text{bm}} + \sum \mathbf{f}_{\text{epi}} / \sum \mathbf{f} $	$< 1.4 \times 10^{-16}$ $10 \varepsilon_{64}$
G10	momentum, motion: free pair, centroid drift	$< 4.4 \times 10^{-15}$ 10^{-12}
G11	control, no plaques: the sheet is not carried	$\lambda \approx 1.25$ vs 3.30 (marginal) 1
G12	standoff equals ℓ_0	$\ell/\ell_0 \in [0.9, 1.1]$ 3.43 — superseded by §2.3: 96% of nodes were unanchored
G13	slip: $\xi = 0$ and $\xi > 0$ must differ	any none — a fixed anchor is a pin; slip IS bond turnover sep- a- ra- tion
<i>the reservoir (§2.4, 05c_remesh, archived 05e)</i>		
G14	refining an UNLOADED sheet changes λ	$< 6.2 \times 10^{-9}$ 10^{-6}
G15	refining a LOADED sheet: λ / area / energy	$(\sqrt{\varepsilon})$ 10^{-12} 2.3×10^{-14}, 0, 0 10^{-9} , 10^{-6}
G16	no hanging nodes	exact 0 bad, 0 rim
G17	mean edge stays in a band as R triples	$[0.8, 107.84]$ (fixed mesh: 3.33) seeded
G18	refinement does not degrade fidelity	≥ 0.9985 vs 0.9873 the fixed mesh's
G18b	mass $\sum_f m_f$ invariant across a split	exact 0, bit-identical
G18c	plaque areal density held at Σ^{-2}	within 0.996 vs control 0.145 10%
G18d	a split does not move the surface: volume, area, centroid	$< 2.6 \times 10^{-16}$, 0, 9×10^{-17} 10^{-12} rel.
G18e	$\rho = m/A$ equals $1/J$	— an identity, not a test
G28	the sheet stays OUTSIDE the epithelium (signed)	radial $+4.3 \times 10^{-4}$ (was -3.6×10^{-3}) gap > 0
G29	no plaque ends up on the wrong side	0% 0.0% (was 89.6%)
<i>before 06: the sheet on a REAL epithelium — numbered from G40 to leave G30–G39 to the adhesion work (05n, 05p, 05q)</i>		
G40	λ^{geo} does not depend on the adhesion's stiffness	κ_n 0.0000 (4.455 at every κ_n) over $8 \times$ moves it $< 5\%$ λ^{geo} 05q within 5% of $\sqrt{A_{\text{ep}}(T)/A_{\text{ep}}(0)}$, NOT of the rs
G41	it tracks a real, dividing surface	

Two conventions make these mean what they say. **The momentum residual is taken on the adhesion's own force pair**, inside the substep, before the drive is added — with the drive's force in the sum it stops being a statement about the reaction. And **a rupture threshold is taken from the load distribution the nominal run measured**, never from a round number: run 127's detach: 0.02 sat above the largest load in its entire run (0.017), so a correctly wired mechanism was reported as a null.

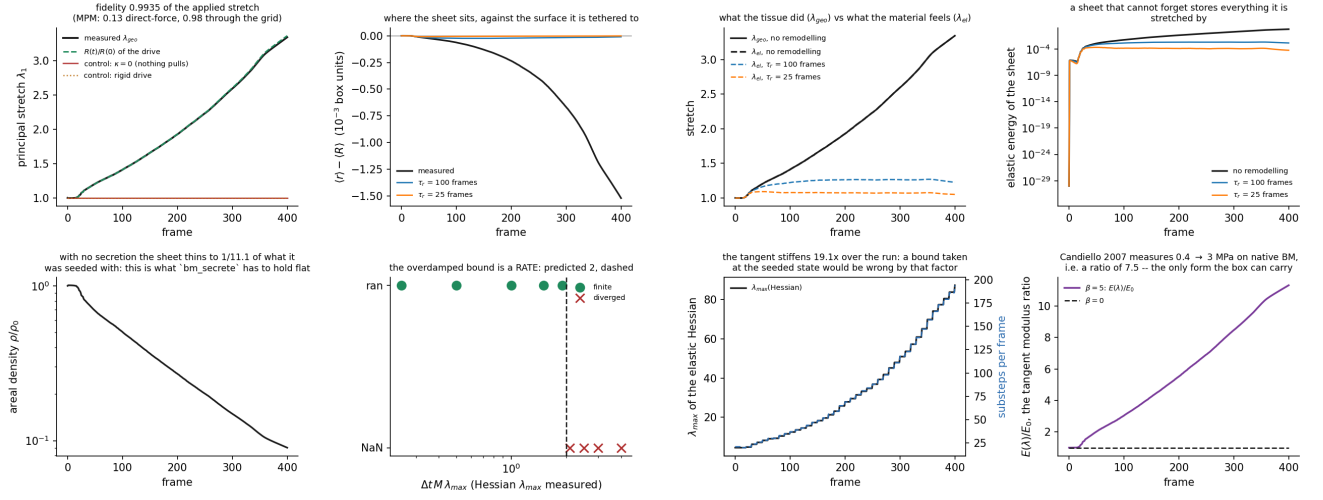


Figure 3: 05a_sheet, the sheet alone on the recorded epithelial surface. **a G5**: the measured λ^{geo} lies on the applied $R(t)/R(0)$ to 0.9935, against 0.13 for the MPM sheet moved by an engine delta; the two flat lines at 1.0000 are **G6** ($\kappa = 0$) and **G7** (a rigid drive), i.e. the measure invents no strain and does not mistake motion for deformation. **b–c** remodelling: at $\tau_r = 25$ frames the standoff improves 1026 $\times$ while λ^{geo} is unmoved — the sheet still goes where the tissue puts it and stops fighting. **f G8**: divergence at the predicted group $\Delta t M \lambda_{\max} = 2$, measured, not assumed. **g** the tangent stiffens 19 $\times$ over the run, which is why the bound has to be re-measured. **h** the stiffening law as a ratio, since MPa is not a quantity this box has.

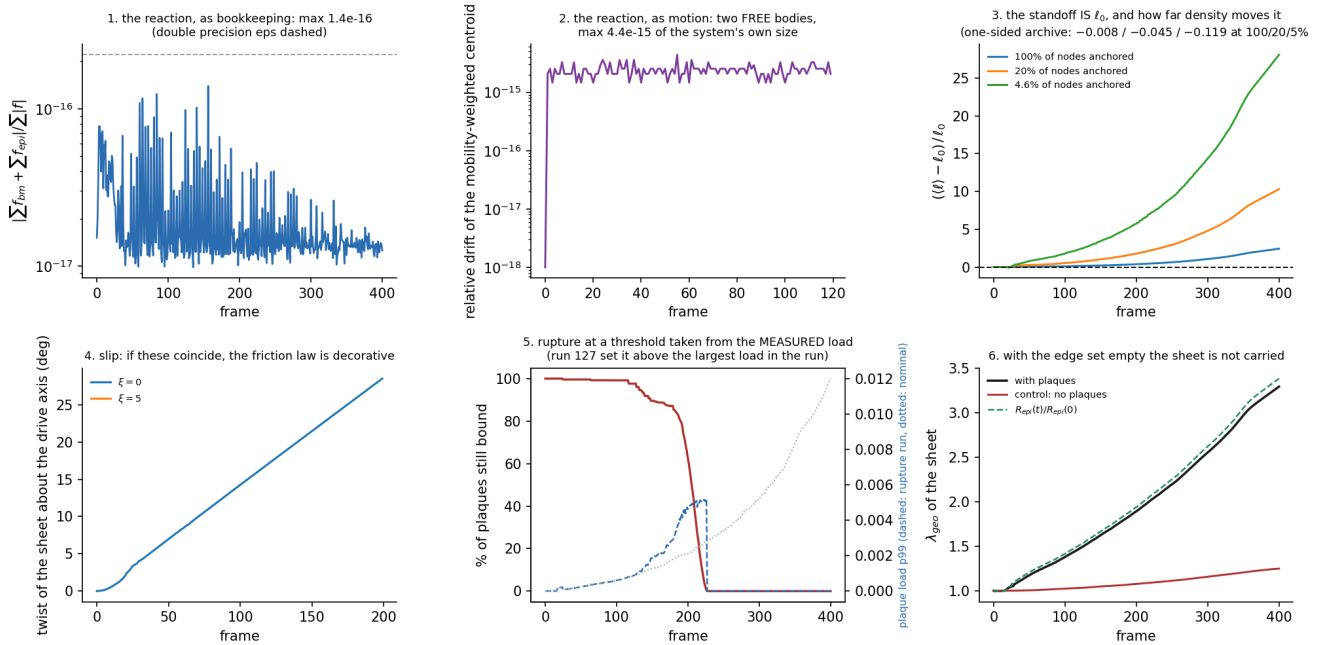


Figure 4: 05b_plaque, the adhesion as an edge set. **1 G9**: the residual on the adhesion's own force pair sits at 1.4×10^{-16} , i.e. at double-precision ε — one operator, a delta to both endpoints, and the reaction cannot be half-implemented. **2 G10**: the same claim as motion, with two free bodies whose mobility-weighted centroid drifts by 4.4×10^{-15} of the system's own size. **3 G12**: the standoff is $3.43 \ell_0$ and not ℓ_0 . **4 G13**: $\xi = 0$ and $\xi > 0$ do not separate, because a fixed barycentric anchor is a tangential pin. **5** rupture is a clock rather than a load, for want of turnover. §4.2 is what each of those three costs, and §2.3 is the scheme that replaces them.

bm_refine

a Divide operator on the bm_face hyperedge set
NUMERICAL MAINTENANCE ONLY: it changes the triangulation and nothing else

$$\mathbf{D}_m^{\text{child}} = \mathbf{D}_m^{\text{parent}} \mathbf{S}_k, \quad \det \mathbf{S}_k = 1/4$$

$$A^0 \rightarrow A^0/4, \quad m_f \rightarrow m_f/4, \quad \mathbf{C}_0, Y_2 \text{ inherited}$$

$$\rho_f = \frac{m_f}{A_f(\text{now})}$$

$$N_{\text{plaque}} = A(t) / \Sigma^2$$

a split is invisible to the material: the child inherits its parent's reference frame, so λ is unchanged, and takes a quarter of its MASS, so Σm_f is unchanged. ρ is derived, never stored, so the dilution term of the mass balance is not integrated -- it is what m/A does on its own.

$\Sigma = 7T, \quad t_0 = 0.3T, \quad T = 0.002$ box units
refine when $(t) > \Sigma/3 = 0.004667$ box
reservoir: 81920 face slots, 40962 node slots

Kanchanawong et al. 2010 Nature 468:580
Rivara 1984 Int J Numer Methods Eng 20:745
engine.py:453 -- 'grow_reserve', the same pattern for MPM

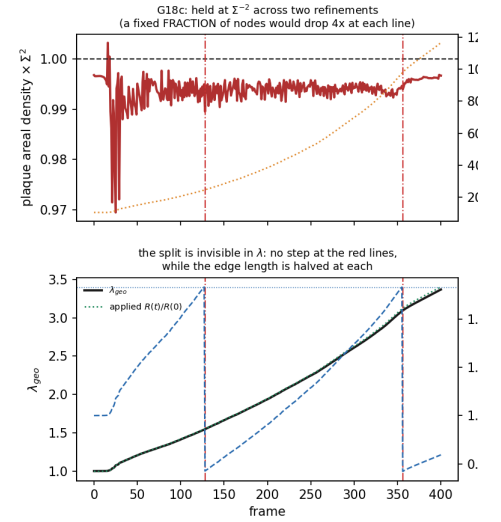
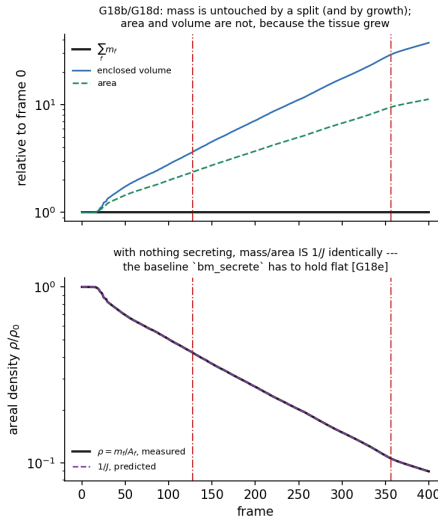


Figure 5: **bm_refine** and what it must carry (05e_conserve). Red dash-dot lines are the two refinements, at frames 131 and 353. **Top left G18b, G18d**: $\Sigma_f m_f$ is flat through both splits and through an area that grows $\times 11$ — bit-identical — while the volume and area grow because the tissue did. **Top right G18c**: the plaque areal density is held at 0.996 of Σ^{-2} ; the fixed-fraction rule 05b used ends at 0.145 . **Bottom left G18e**: with nothing secreting, m/A IS $1/J$ identically — an identity, drawn because it is the baseline **bm_secrete** has to hold flat. **Bottom right** the split is invisible in λ (no step at the red lines) while the mean edge length is halved at each.

bm_secrete

a Lateral operator on the bm_face hyperedge set
(it was specified as a Divide, with mass as a scalar on a face, no new face is needed)

$$\frac{dm_f}{dt} = s(\mathbf{x}, t) A_f - \frac{m_f}{\tau_{bm}}$$

$$\rho_f = \frac{m_f}{A_f(\text{now})}$$

$$s^* = \rho_0 \left(\frac{1}{\tau_{bm}} + \dot{\lambda} \right)$$

a face dies where $\rho_f < \rho_{\text{crit}}$

the dilution term is NOT integrated: it is what m/A does on its own as the tissue grows. Adding it as well would count it twice. Mass goes onto faces that already are the sheet, so run 128's second shell in the lumen cannot happen here.

$\tau_{bm} = 40$ frames ($t_{10} = 3-10$ h = $13-93$ frames)
 $\rho_{\text{crit}} = 0.35 \rho_0$ $\rho_{\text{target}} = \rho_0$
Matsubayashi et al. 2020 Dev Cell 54:33
Yurchenco 2011 Cold Spring Harb Perspect Biol 3:a004911
run 128: the archived secretion that missed the sheet

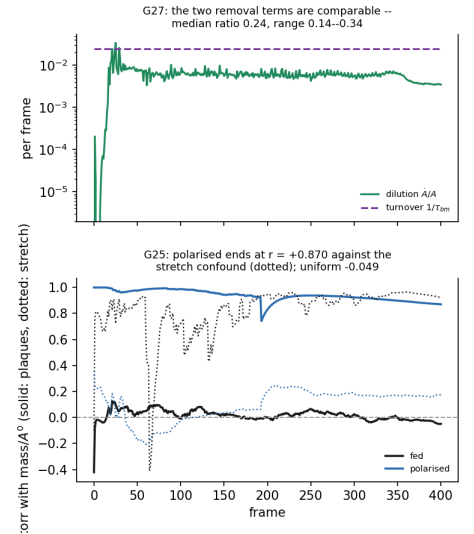
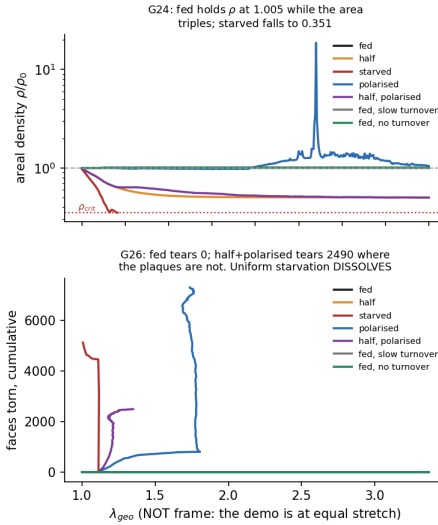


Figure 6: **bm_secrete** (05f_secrete), in the register of §3's operator table: the operator names itself, states the equations it is, and every panel is a measurement that could have come back wrong. **Top left G24**: fed holds ρ at 1.005 while the area triples, against 0.351 starved. **Top right G27**: the two removal terms are *not* the same size — dilution is 0.238 of turnover at $\tau_{bm} = 40$ frames. **Bottom left G26**, plotted against λ and not against the frame, because “starved tears and fed does not” is a statement about supply only if the two are compared at the same deformation. **Bottom right G25**: polarised deposition ends at $r = +0.88$ with the plaques against uniform's -0.05 , with the stretch correlation drawn beside it as the control that separates deposition from mechanics.

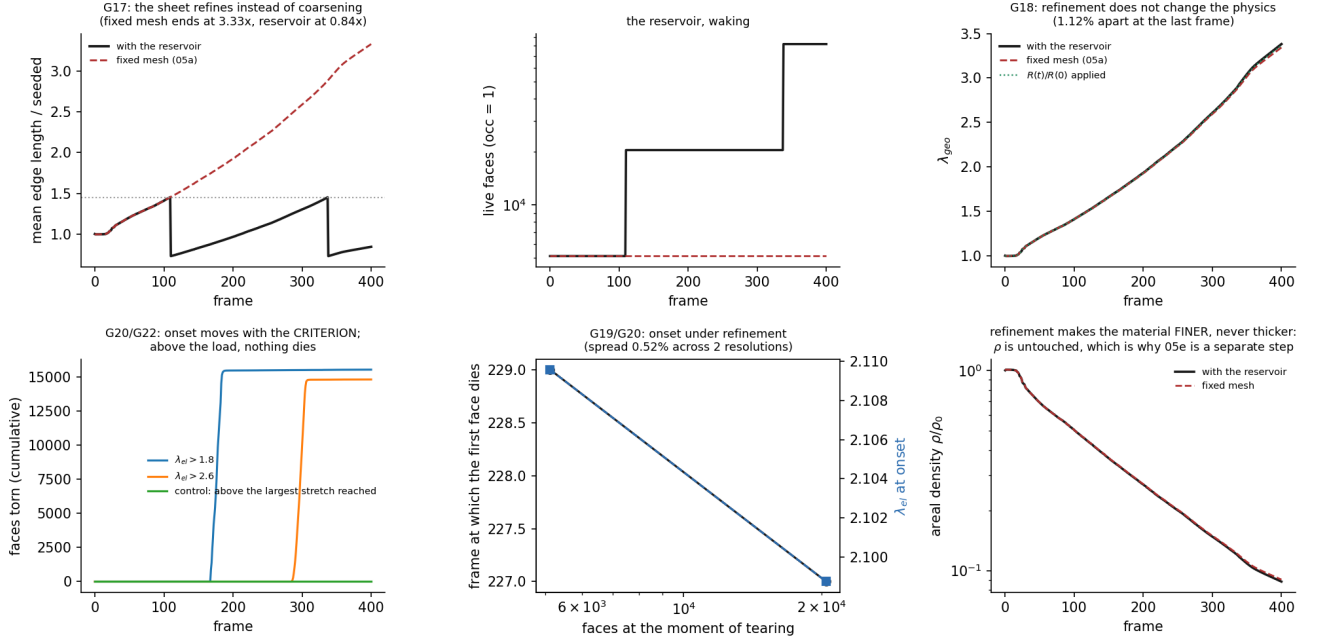


Figure 7: 05d_tear: tearing, and the gate the MPM sheet was never given. **Bottom middle G19**: the same criterion at 5,115 and 20,471 faces tears at $\lambda^{el} = 2.110$ and 2.099 — 0.52% apart, against a 5% threshold — so the tear is set by the material and not by the discretisation. **Bottom left G20, G22**: onset moves with the criterion (frames 167 / 227 / 285 at thresholds 1.8 / 2.2 / 2.6) and a criterion above the largest stretch reached kills 0 of 81,920 faces — which is the run-127 control, run on purpose. **Top row G17, G18**: the reservoir holds the edge length at 0.84 of seeded where a fixed mesh reaches 3.33, and refinement improves the fidelity rather than merely preserving it.

4.1 The sweeps behind the table

05a_sheet — the sheet alone, tethered to the recorded epithelial surface (smap: 32×64 directions $\times$ 402 frames, R going $0.0398 \rightarrow 0.1351$ box units, a ratio of 3.387), 2562 nodes, 5120 faces, no matrix and no plaque.

run	τ_r	λ^{geo}	λ^{el}	standoff	what it says
nominal	∞	3.344	3.344	-1.52×10^{-3}	the sheet stores all of it
tau100	100 fr	3.384	1.226	-9.96×10^{-6}	153 $\times$ better
tau25	25 fr	3.384	1.050	-1.48×10^{-6}	1026$\times$ better
beta5	∞	3.063	3.063	-1.29×10^{-2}	stiffening makes it worse
control $\kappa = 0$	—	1.0000	1.0000	—	the measure invents nothing
control rigid	—	1.0000	1.0000	—	motion is not deformation

The third row is a mechanism, not a tuning. Without remodelling the sheet’s own hoop stress pulls it inward faster than a stable tether can hold it out — which is what `HANDOVER.md` concluded from runs 149–155 and could not test, because `membrane_tau` was 0 in every one of them. Turning it on moves the standoff by three orders of magnitude while λ^{geo} stays at 3.38: the sheet still goes where the tissue puts it, it simply stops fighting. $\beta = 5$ is the same statement from the other side — a sheet that stiffens as it stretches fights harder, and its standoff is $8.5\times$ worse at $E(\lambda)/E_0 = 11.3$.

05b_plaque — 2562 membrane nodes against a 642-vertex epithelium, 2562 plaques, $\ell_0 = 0.3T$. G9 and G10 pass at machine precision, which is what this run exists for: the reaction is present, and present as motion and not only as bookkeeping. The density sweep, against the archived one-sided tether:

anchored	spacing	standoff, two-sided plaque	standoff, one-sided tether (archived)
100%	$1.5T$	$+1.46 \times 10^{-3}$	-8.2×10^{-3}
20%	$3.4T$	$+6.19 \times 10^{-3}$	-4.5×10^{-2}
4.6%	$7T$ (the measured Σ)	$+1.68 \times 10^{-2}$	-1.19×10^{-1}

At this model’s plaque density — which is NOT the density the literature measures, see §2 — which on this mesh is 4.6% of the nodes — the two-sided plaque sits $7\times$ closer to the surface than the one-sided tether did. The sign has flipped, and that is §4.2.

05c_remesh — the same sheet with two levels of reserve (81,920 face slots, 40,962 node slots), against the identical run with the reserve switched off. Refinement fires twice, at frames 110 and 338, each time when the mean edge reaches $1.45\times$ its seeded value: $5120 \rightarrow 20480 \rightarrow 81920$ faces, $2562 \rightarrow 10242 \rightarrow 40962$ nodes.

	faces	mean edge / seeded	λ^{geo}	fidelity vs applied 3.387	standoff
reservoir, 2 levels	81,920	0.84	3.3817	0.9985	-9.89×10^{-5}
fixed mesh (05a)	5120	3.33	3.3440	0.9873	-1.52×10^{-3}

G17 passes. G18 was written as “the refined run must match the fixed-mesh run to 1%” and the two are 1.12% apart — **which is the gate being wrong, not the run**. The fixed mesh is not the truth: a coarse triangle chords a curved, stretching surface and under-reports how much it stretched, so refinement moves the fidelity from 0.9873 to 0.9985 and the residual error from 1.3% to 0.15%. The gate is therefore restated against the *applied* stretch, where refinement can only help, and G18 as originally phrased would have rejected an improvement. The standoff improves by $15\times$ in the same run, for the same reason: more nodes tethered to the surface track it more closely.

05e_conserve — the reservoir and the plaque in one rig, so that a refinement has something to break, with mass as the state (Eq. 10) and **plaque_seed** holding Σ^{-2} . Over 401 frames it refines twice, at frames 131 and 353, and the sheet’s area goes $\times 11$:

	threshold	measured	
G18b	$\sum_f m_f$ across a split	exact	0 , and 0 over the whole run
G18c	plaque density $\times \Sigma^2$	within 10%	0.996 (104 $\rightarrow$ 1056 plaques)
G18c	<i>control</i> : 05b’s fixed-fraction rule	—	0.145
G18d	volume / area / centroid across a split	$< 10^{-12}$	2.6×10^{-16} , 0 , 9.1×10^{-17}
G18e	$\rho = m/A$ against $1/J$	—	1.4×10^{-16} : an identity, not a test

Two things in that table are worth saying in words. **Mass is bit-identical across a split and across the whole run**, which is what licenses **bm_secrete** to be built on top: a substrate that lost material when it refined could not tell secretion from arithmetic. And **the control is a larger failure than predicted**: 05b’s rule loses a factor of 6.9, not the factor of 4 per refinement one would guess, because a fixed fraction is wrong at each split *and* continuously as the area grows between them.

One operator changed kind because of this. **bm_secrete** was written as a *Divide* — wake reserve faces to add material, which is what a particle model must do. With mass carried as a scalar on a face it is a *Lateral* operator: it raises m_f on faces that already exist, and the reservoir is needed for resolution and not for supply. That is a simplification the ρ -as-state form would have hidden.

4.2 Three gates that failed, and none of them is a parameter

The standoff is $3.43\ell_0$, not ℓ_0 (G12). The sheet lags the expanding epithelium, so every plaque is *extended* rather than sitting at its rest length, and the extension is what carries the sheet outward. ℓ_0 therefore sets the standoff only in a quasi-static sheet; in a growing one it is ℓ_0 plus a lag, and the lag is the same hoop stress §4.1 isolates. The prediction is sharp and is the next run: 05b with $\tau_r = 25$ should collapse $3.43 \rightarrow \sim 1$.

Friction is currently unmeasurable in this rig (G13). With $\xi = 0$ the sheet twisted 28.49° against a drive of 28.50° — it followed the epithelium essentially exactly with no friction at all. The reason is structural: a plaque anchored to a *fixed* barycentric point is a pin in the tangential direction, because that anchor is a material coordinate that rotates with the face. ξ can only set how fast the sheet equilibrates to a pin, never whether it slides. **Slip requires the anchor to move over the epithelial surface**, i.e. **plaque_seed** as turnover. §5 of **note_spheroid_bm_ecm** asked for friction where what was needed was turnover; that is a correction to the design, and the run is what found it.

Turnover does not merely match growth — it dominates it, and §9 of note_spheroid_bm_ecm says otherwise. That note argues the two removal terms of the mass balance are the same size, “turnover is not a correction to growth; it is the same size as growth, which is why the sheet can accommodate it at all”. Measured on the geometry rather than estimated: $\dot{A}/A$ has median 5.96×10^{-3} per frame against $1/\tau_{\text{bm}} = 2.5 \times 10^{-2}$, a ratio of **0.238** — turnover runs $4.2\times$ faster than dilution at the mid-range half-life, and $1.8\times$ faster (0.555) at $\tau_{\text{bm}} = 93$ frames, the 10 h end. The claim therefore holds only near the slow end of the literature’s 3–10 h range, and the correct statement is that *turnover dominates growth by a factor of two to four*. It is the same conclusion for the sheet — a membrane that replaces itself faster than it is diluted can hold its density — reached by a different margin than the note asserts.

Rupture is all-or-nothing for a structural reason. At a threshold of 0.0052, inside the measured load range

(the nominal's p99 reaches 0.0120), *every* plaque broke. With a load that rises monotonically as the tissue grows and nothing that re-forms a bond, any threshold inside the range is reached by every plaque eventually: the fraction bound is a function of time and not of load. A partial-rupture regime needs a distribution of thresholds across plaques, or turnover. Reporting a fraction bound without one of those would be reporting a clock.

4.3 What this licenses, and what it does not

Licensed: the sheet's deformation is measured rather than advected (G1–G5); its stability limit is a rate that is measured rather than asserted (G8); its adhesion is a relation that returns its reaction to machine precision (G9, G10); and its *size* is now a state variable that can be changed without disturbing the mechanics at all (G14–G16, with area and energy bit-identical across a split). Those are the four things the archived line could not get, and they are what 04 needs before a membrane can be put into it.

Not licensed: the standoff is not yet ℓ_0 ; slip needs turnover, not friction; rupture needs a threshold distribution; there is no secretion, so Eq. (9) is a baseline ($\rho \rightarrow 1/11.1$) and not yet a mechanism; refinement is global rather than adaptive, and a crack tip will want local; there is no coarsening, so the sheet can only get finer; there is no bending stiffness, so it cannot fold, and a hole's rim has no line tension; there is no matrix and no **fibril**, so nothing yet removes the epithelium's direct push on the stroma; and the epithelium in 05b is a driven icosphere, not the vertex model — **mesh_contact** certifies that interface separately, and joining the two is what 05g does.

Exit gates: what must be true before 06

The sheet is not ready to be joined to the matrix and the vertex model because of what it has passed; it is ready when the following are closed. These are stated *before* the runs that test them, and each names the consequence of failing rather than a threshold to be relaxed afterwards. Three of them have never been run at all, which is a stronger reason to hold than any of the marginal results above.

gate	threshold	status
<i>A — never run. The scheme rests on these and none has been tested.</i>		
G59 the density adhesion reproduces the discrete one where the discrete one is valid: raise the plaque density until there is more than one cluster per node and the two models must agree	standoff and traction within 5%	05j
G60 the field integral $\sum_f c_f A_f$ is invariant across a 1→4 split	exact	05k
G61 protease on a torn face is <i>accounted</i> , not silently dropped	the balance closes	05k
G62 the steady field profile is mesh-independent	< 5% across one refinement	05k
<i>B — measured and failing. Close or waive explicitly.</i>		
G42 the field's own length is resolved: $\sqrt{D/k} \geq 2h$	hard	0.05 vs $h = 0.076$ 2.03 — marginal partly: 05h_3 breaches under a UNIFORM source, so it is not a stencil; the radius has never been measured
G55 the bell peaks at $c_T = K$	within 2×	
G46 the breach is <i>larger</i> than its source, and the excess tracks $\sqrt{D/k}$	radius ratio > 1	
G63 the sheet sits <i>across</i> the bell, not on its shoulder	band fraction > 25%	0.7–2.6% in every 05h run
<i>C — required by 06 itself, and never attempted.</i>		
G64 every operator runs <i>simultaneously</i> for the full 401 frames: elasticity, remodelling, contact, adhesion with turnover, secretion, refinement, proteolysis	no divergence, no dissolution	05m
G65 momentum still closes with <i>all</i> couplings summed (adhesion + contact + fibril)	< 10ε	05m
G66 the cost is affordable: wall clock per frame at the final mesh with the matrix present	reported, with the dominant term named	05m

G59 carries a stop condition and it is the one to watch. If a discrete-plaque model at high density does not reproduce the bond-density model, then the density is not a coarse-graining of the discrete picture — it is a different model wearing its name, and the whole of §2.3 would have to be withdrawn rather than patched. Every other gate here can be waived with a stated reason; that one cannot.

5. References

References

- [1] Karagiannis, E.D., Popel, A.S. (2004) A theoretical model of type I collagen proteolysis by matrix metalloproteinase (MMP) 2 and membrane type 1 MMP in the presence of tissue inhibitor of metalloproteinase 2. *J. Biol. Chem.* **279**(37):39105–39114. — the canonical ODE network for MT1-MMP / TIMP-2 / proMMP-2 / MMP-2, and the source of the reaction topology used here.
- [2] Sato, H., Takino, T. (2010) Coordinate action of membrane-type matrix metalloproteinase-1 (MT1-MMP) and MMP-2 enhances pericellular proteolysis and invasion. *Cancer Sci.* **101**(4):843–847.
- [3] Activation of MMP-2 by MT1-MMP through an artificial receptor for proMMP-2 generates active MMP-2. *Cancer Res.* (2008) **68**(21):9096. — the dissection showing the descending limb is MT1-MMP inhibition and not receptor saturation.
- [4] Bonet, J., Wood, R.D. (2008) *Nonlinear Continuum Mechanics for Finite Element Analysis*, 2nd ed. Cambridge University Press. — the constant-strain-triangle membrane of §3.
- [5] Candiello, J., Balasubramani, M., Schreiber, E.M., Cole, G.J., Mayer, U., Halfter, W., Lin, H. (2007) Biomechanical properties of native basement membranes. *FEBS J.* **274**(11):2897–2908.
- [6] Bell, G.I. (1978) Models for the specific adhesion of cells to cells. *Science* **200**(4342):618–627. — the force-dependent off-rate $k_{\text{off}} = k_{\text{off}}^0 e^{f/f_b}$ that makes rupture a load and not a clock.
- [7] Changede, R., Sheetz, M. (2017) Integrin and cadherin clusters: a robust way to organize adhesions for cell mechanics. *BioEssays* **39**(1):1600123. — nascent adhesions are ~ 100 nm and contain ~ 50 integrins; large adhesions are loose aggregates of such clusters.
- [8] *Differential spatial regulation and activation of integrin nanoclusters inside focal adhesions.* eLife reviewed preprint 105270. — STORM: nanocluster median size 35–40 nm, centre-to-centre ~ 555 nm, spacing set cell-intrinsically. Cited as a preprint.
- [9] Kanchanawong, P., Shtengel, G., Pasapera, A.M., Ramko, E.B., Davidson, M.W., Hess, H.F., Waterman, C.M. (2010) Nanoscale architecture of integrin-based cell adhesions. *Nature* **468**:580–584. — iPALM, and it measures the VERTICAL architecture: a ~ 40 nm integrin-to-actin core, actin peaking at $z = 97$ nm. It contains no lateral plaque diameter and no lateral spacing, so it licenses ℓ_0 and NOT Σ .
- [10] Matsubayashi, Y., Sánchez-Sánchez, B.J., Marcotti, S., Serna-Morales, E., Dragu, A., Díaz-de-la-Loza, M.-C., Vizcay-Barrena, G., Fleck, R.A., Stramer, B.M. (2020) Rapid homeostatic turnover of embryonic ECM during tissue morphogenesis. *Dev. Cell* **54**(1):33–42.
- [11] Kelley, L.C., Chi, Q., Cáceres, R., Hastie, E., Schindler, A.J., Jiang, Y., Matus, D.Q., Plastino, J., Sherwood, D.R. (2019) Adaptive F-actin polymerization and localized ATP production drive basement membrane invasion in the absence of MMPs. *Dev. Cell* **48**(3):313–328.
- [12] Yurchenco, P.D. (2011) Basement membranes: cell scaffoldings and signaling platforms. *Cold Spring Harb. Perspect. Biol.* **3**(2):a004911.
- [13] Keene, D.R., Sakai, L.Y., Lunstrum, G.P., Morris, N.P., Burgeson, R.E. (1987) Type VII collagen forms an extended network of anchoring fibrils. *J. Cell Biol.* **104**(3):611–621.
- [14] Rivara, M.-C. (1984) Algorithms for refining triangular grids suitable for adaptive and multigrid techniques. *Int. J. Numer. Methods Eng.* **20**(4):745–756. — the local alternative to the global $1 \rightarrow 4$ split of §2.4, for when a crack tip needs it.
- [15] Okuda, S., Inoue, Y., Eiraku, M., Sasai, Y., Adachi, T. (2013) Reversible network reconnection model for simulating large deformation in dynamic tissue morphogenesis. *Biomech. Model. Mechanobiol.* **12**(4):627–644.
- [16] Holzapfel, G.A., Gasser, T.C., Ogden, R.W. (2000) A new constitutive framework for arterial wall mechanics and a comparative study of material models. *J. Elasticity* **61**:1–48.